

MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
 243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Automotive Transmission		
Course Code:	PC-AUE601	Category:	Professional Core Courses
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Theory of Machine and Design of Machine Element
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	This course provides the knowledge, gear box clutch		
2	Principle of operation and performance of various components and drives in an automotive transmission system.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Clutch: Requirements, Types- Construction, working and operating characteristics of single plate clutch, multi plate clutch, semi centrifugal, centrifugal clutch. Construction and working of cone clutch, electromagnetic clutch, over running clutch. Clutch linkage – mechanical and hydraulic. Clutch energy dissipated. Clutch lining materials. Deriving the equation for torque capacity of a single plate and multi-plate clutch. Problems involving torque capacity and axial force of single plate and multi-plate clutch. Trouble shooting. Service procedure.	6
2	Gearbox: Necessity. Construction and working of sliding mesh, constant mesh, synchromesh gearbox. Over drive mechanism. Gear shift mechanisms. Total resistance to motion- traction and tractive effort - acceleration - calculation of gear ratio for vehicles - design of three speed gear box and four speed gear boxes. Performance characteristics in different speeds. Speed synchronizing devices. Gear materials. Gear lubrication. Transfer case, Problems in gear box involving gear ratios and various gradients and total resistance calculation. Design the clutch and gear box for a given engine power and vehicle load.	7
3	Hydrokinetic Fluid coupling and Torque converter: Introduction to fluid coupling, Fluid coupling - construction and principle of operation, Drag torque and various drag reducing devices, Performance characteristics of fluid coupling, Problems on design and torque capacity of fluid coupling. Torque converter and converter coupling - construction and principle of operation. Torque converter performance terminology, Multistage torque converter - construction and working, Poly phase torque converter - construction and working. Performance characteristic of multistage and poly phase torque converters.	7

4	Automatic Transmission: Relative merits and demerits when compared to conventional transmission, Principle of working of epicyclic gear train, Planetary gear box - construction and working, Automatic gear box consideration, Three speed & reverse transaxle and four speed & reverse longitudinal mounted automatic transmission mechanical power flow, Fundamentals of a hydraulic control system and basic principle of a hydraulic control gear shift mechanism for automatic transmission, Electronic-hydraulic control automatic transmission, continuously variable transmission, semiautomatic transmission system. Problems in automatic transmission involving gear ratio.	7
5	Hydrostatic Drives and Electric Drives: Introduction to hydrostatic drives, Working principle and types of hydro static drives, Advantages and limitations of Hydrostatic drive, Comparison of hydrostatic drive with hydro dynamic drive, Construction and working of Janny Hydrostatic drive. Introduction to Electric drive and Layout of Electric drive, Principle of Early Ward Leonard control system of electric drive, Principle of Modified Ward Leonard control system of electric drive, Advantages, limitations and performance characteristics of electric drive.	6
6	Different Automatic Transmission: Automated Manual transmission, S-Tronic transmission, Dual Clutch Transmission, Direct shift gear box (DSG), Tiptronic transmission, Electronic control transmission integrated intelligent control system (ECTi).	3
Total		36

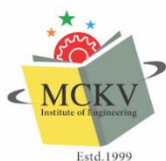
Course Outcomes:

After completion of the course, students will be able to:

1	Explain the constructional, working principle and performance of various types of manual, semi-automatic and automatic transmission of an automobile.
2	Gain knowledge about various hydrodynamic drives.
3	Classify the principle of operation and performance of various hydrostatic and electric drives.
4	Design a gear box for a given engine power and vehicle load.

Learning Resources:

1	Giri N.K., Automobile Mechanics, Khanna Publishers, New Delhi
2	Crouse W.H. and Anglin D.L., Automotive Transmission and Power Train Construction, McGraw Hill.
3	Naunheimer H., Bertsche B., Ryborz J. and Novak W., Automotive Transmission: Fundamentals, Selection, Design and Application, 2nd Edition, Springer, 2011.
4	Heldt P.M., Torque Converters, Chilton Book Co.
5	Newton S. and Garrot, Motor Vehicles, SAE International and Butterworth Heinemann, 2001.
6	CDX Automotive, Fundamentals of Automotive Technology, Principles and Practice, Jones &Barlett Publishers, 2013.
7	SAE Transactions 900550 & 930910.
8	Heinz H., Advance vehicle Technology, Butterworth-Heinemann



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Course Name:	Automotive Electrical and Electronic Systems		
Course Code:	PC-AUE 602	Category:	Professional Core courses
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic Concept of Electrical & Electronics Engg., Sensors, Transducers.
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To acquire the knowledge of Electrical and Electronics engineering concepts.		
2	To learn, identify, demonstrate and compare the various components of different internal and external parts of a vehicle.		
3	To understand the advance technologies used in vehicles.		
4	To learn the role of various sensors, transducers in vehicle management system and its effects in enhancing safety of the vehicle and nature.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	D.C Machine: Main construction Features, Armature Winding, Commutator, Basic principle of d.c generator, Slip-ring, Commutation, Operating characteristics and application of dc generator, armature reaction, total loss in dc generation, working principle of dc motor, Types of dc motor and its characteristics, speed control of dc motor, Three point starter- basic construction and working principle. Principle of transformer and its application, Stepper- motor, Different types of relay.	8
2	Transducer & Sensors: Introduction, Mechanical spring devices, Resistive transducer, Capacitive Transducer, Strain gauges, Thermistor, Thermocouples, LVDT, RVDT, Photoelectric transducer, Digital displacement transducer, Oxygen Sensors, Position Sensors-Fuel metering/vehicle speed sensor and detonation sensor- Altitude sensor, Flow sensor, Solenoids.	5
3	Starting system: Condition of starting, Behavior of starter during starting, and its characteristics, Principle and construction of starter motor, working of different starter device units. Starter Switches.	4
4	Electronic Fuel Injection and Ignition System: Introduction, Feedback carburetor system, Multi port injection, Ignition system controls, Electronic Ignition System, Advantage of Electronic Ignition System., Solid-State ignition system.	5
5	Automotive Electronics: Transfer function, Application of control system in automobile industry, open-loop system and closed-loop system, vehicle motion control, Review of fundamentals of digital electronics,	8

	Logic gates and their operations, Combinational & Sequential Logic Circuits, Signal processing devices, Data conversion devices, Introduction to microcontroller, Various Microcontroller based operation in a vehicle.	
6	Lighting system & Accessories: Details of head light and side light, Head light aiming, Speedometer, Odometer, Horn, Horn-Relay, Wiper system, Trafficator. Digital Engine control System: Engine cranking and warm up control, Acceleration enrichment, Integrated engine control system, Exhaust emission control engineering.	6
Total		36

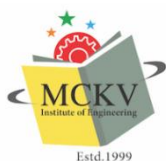
Course Outcomes:

After completion of the course, students will be able to:

1	Understand the basic terminologies, components and concepts of electrical and electronics engineering.
2	Explain the working components and mechanism of different internal and external parts of a vehicle.
3	Explain the purpose, circuits, construction and working of components of starting system, ignition system, lighting and accessories systems.
4	Understand the role of various sensors, transducers in advanced vehicle management system and its influence in controlling pollution, enhancing safety of the vehicle.

Learning Resources:

1	Automobiles Electrical Equipment, Young, A. P., & Griffiths. L., English Language Book Society & New Press, 1990.
2	Automotive Electrical Equipment, Kholi. P. L, Tata McGraw-Hill Co. Ltd. New Delhi, 1975.
3	Electrical Machine, J.B. Gupta.
4	Automotive hand Book, Robert Bosch, Bently Publishers, 1997.
5	Fundamentals of Digital Circuits, A.Anand Kumar , 3rd, PHI
6	The 8051 Microcontroller, Kenneth J. Ayala, 1996, Penram International Publishing



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Course Name:	Hybrid and Electric Vehicles		
Course Code:	PE-AUE601A	Category:	Professional Elective Courses
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic Electrical and Electronics and Automotive Engine and Chassis
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To study the characteristics behaviour of engine, electric motor, controller and energy storage devices for vehicle propulsion systems		
2	To study and analyze the renewable energy systems to develop zero emission vehicles.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction to Hybrid Electric Vehicles: History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies.	3
2	Conventional Vehicles: Basics of vehicle performance, vehicle power source characterization, transmission characteristics, mathematical models to describe vehicle performance.	4
3	Hybrid Electric Drive-trains: Basic concept of hybrid traction, introduction to various hybrid drive-train topologies, power flow control in hybrid drive-train topologies, fuel efficiency analysis.	4
4	Electric Drive-trains: Basic concept of electric traction, introduction to various electric drive-train topologies, power flow control in electric drive-train topologies, fuel efficiency analysis.	4
5	Electric Propulsion unit: Introduction to electric components used in hybrid and electric vehicles, Configuration and control of different motors drives like DC motor drives, Induction Motor drives etc.	5
6	Energy Storage: Introduction to Energy Storage Requirements in Hybrid and Electric Vehicles; Electrochemical Batteries, Lead-Acid Batteries, Nickel Based Batteries, Lithium Based Batteries, Ultra Capacitors- Basic Principles and Performance, Ultrahigh-speed flywheels- Basic Principle and Power Capacity, Fly Wheel technologies. Energy recovery by regenerative braking.	5
7	Sizing the drive system: Matching the electric machine and the internal combustion engine (ICE), Sizing the propulsion motor, sizing the power electronics, selecting the energy storage technology, Communications, supporting subsystems.	3

8	Case Studies: Design of a Hybrid Electric Vehicle (HEV), Design of a Battery Electric Vehicle (BEV).	4
9	Fuel Cell Electric Vehicles: Operating principles of fuel cells, Fuel and oxidant consumption, Fuel cell system characteristics, Fuel cell technologies- Proton Exchange membrane fuel cells, Alkaline Fuel cells, Phosphoric acid fuel cells, Molten carbonate fuel cells, Solid oxide fuel cells, Fuel supply- Hydrogen Storage-Hydrogen production, Ammonia as hydrogen carrier, Non-Hydrogen fuel cells, Fuel Cell Hybrid Vehicle Drive Train.	4
Total		36

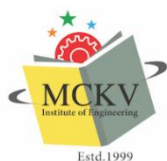
Course Outcomes:

After completion of the course, students will be able to:

1	Choose a suitable drive scheme for developing an electric hybrid vehicle depending on resources.
2	Choose proper energy storage systems for vehicle applications.
3	Design and develop basic schemes of electric vehicles and hybrid electric vehicles.

Learning Resources:

1	Husain I., Electric and Hybrid Vehicles: Design Fundamentals, CRC Press.
2	Ehsani M., Gao Y., Gay S.E. and Emadi A., Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press.
3	Larminie J. and Lowry J., Electric Vehicle Technology Explained, Wiley.
4	John M. Miller, Propulsion Systems for Hybrid Vehicles, The Institution of Engineering and Technology
5	Chris Mi, M.Abul Masrur and David Wenzhong Gao, 1st Edition, Hybrid Electric Vehicles, John Wiley & Sons, Ltd, 2011



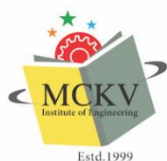
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Course Name:	Transport Management & Motor Vehicles Act		
Course Code:	PE-AUE601B	Category:	Professional Elective Courses
Semester:	Sixth	Credit:	03
L-T-P:	3-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	Focus on fleet policy development, safety programs initiatives, vehicle selector creation based on exhaustive lifecycle cost analyses, implementation of a corporate sustainability program, and the establishment of metrics to manage fleet suppliers.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Transit Operation: Modes of transport, road transport - Types of roads, advantages, motor transport in India, Route planning - Route location, stop location, route schedules, vehicle and labor scheduling. Traffic control - Traffic signals, signal timing.	7
2	Costs & Fares: Operating costs and types of vehicles - types of fare structure, types of fare collecting methods - Requirement of buses and frequency, construction of bus station.	7
3	Forms of Ownership: Sole proprietorship, partnership, private limited company, public limited company, statutory company, local authority undertaking / municipal transport company, joint venture.	7
4	Garage Management and Vehicle Maintenance: Garage administration, types of garages, break down truck symbol, government approved workshops – Tools- Objectives of maintenance, breakdown maintenance, preventive maintenance, tyre maintenance tips and failures. Fuel saving techniques and fitness certificate.	7
5	Legal Aspects: Motor Vehicle Act 1988, Registration, necessity of permits, insurance, test of competence to drive, mistake / offences for which a driver can be punished, adult workers - Hours of work, running time, split duty, journey time, round journey time, layover, frequency.	8
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	Understand science and technology behind current and future modes of transportation.	
2	Apply economic principles to management decisions	
3	Comprehend commercial vehicle regulations and legal knowledge along with a practical understanding of key areas	

Learning Resources:	
1	Government Publication, "The Motor Vehicle Act ", 1988.
2	Doke J., Fleet Management, McGraw Hill Co.,1984.
3	Crouse, William H, Anglin, Donald L, “Automotive Mechanics”, McGraw-Hill Companies, 2007.
4	Kitchin L.D, "Bus operation ", Illiffe and Sons Ltd., London, 1992.
5	Gilles, Tim, “Automotive Service – Inspection, Maintenance, and Repair”, Alar Elken Publications, 2007.
6	Bhandarkar S.L., Vehicle Transport Management, Dhanpat Rai & Co., 2016.
7	Gupta R.B., Transport Management, Satya Prakashan, 2016
8	White P.R., Public Transport: Its Planning, Management and Operation, Natural and Built Environment Series, Kindle Edition, September 2008.



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Course Name:	Tyre Technology		
Course Code:	PE-AUE601C	Category:	Professional Elective Courses
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Automotive Body & Chassis Engineering
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To understand various components used and their function of tyres.		
2	To design and suitable compounding formulation for various tyre components.		
3	To know the building & curing of tyres.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	INTRODUCTION: The current status of tyre industry in India and its future prospectus. Tyre sizing and marking on the tyres. Different types of tyres – bias, bias belted radial, tube type and tubeless tyres their basic features and performance comparison. Different components of a tyre, its geometry, basic functions. Functions of a pneumatic tyre – load carrying, vibration and noise reduction, the tyre function as a spring, contribution to driving control and road adhesion, the tyre friction contribution to driving control, steering control and self aligning torque.	7
2	FABRIC PREPARATION and CALENDERING: Fabrics of the Tyre Industry: Cotton, Rayon, Nylon & steel cords – manufacture, construction – styles and presentations. Bonding methods – Fabric bonding necessities of stronger fabrics leading to bonding methods developments. Wet & dry bonding systems – dip and hot stretch process for Nylon. REL-VP latex systems – and parameters for dip & hot stretch process for Nylon. Modified surface treatment needed for polyesters & glass fabric - Metal coating for steel cord. Radical Tyre fabrics – Aromatic Nylon (Kevlar) and other special fabric reinforcement systems and their use - Testing of dipped fabrics. Calendering process, Skimming & frictioning process preparation of bead wrapper and chaffer-on fabrics on 3 roll calenders. Topping process on calendar - Limitation of 3 roll calenders and advantages of 4 roll calenders-process control aspects – economics - Relation between ends per inch and calendering process. Inner, outer and breaker fabrics. Compound fabric ratios and compound design consideration for different styles of fabrics - Defects of calendered fabrics and their remedies.	8
3	THREAD EXTRUSION AND BEAD CONSTRUCTION Basic concepts of Extrusion. Die swell & shrinkage phenomenon – effect of compounding parameters on these phenomena. Die design and	7

	theoretical calculation of tread weight. Effect of viscosity & temperature on extrusion. Extruded tread profile – critical dimensions. Dual extruder – Cap & base concept relation to tyre wear parameters like tread wear heat buildup etc. Cross head extruder wire coating process - Bias cutting and pocket making: Bias angle specification and the significance Horizontal and vertical laying of coated wore. Apex preparation on extruder and profile calender Bead wrapping and flipping operations. Single and double bead concept and preliminary calculation of bead safety factors. Width and angle adjustments splicing and identification.	
4	TYRE BUILDING Tyre building inputs: Inner liners, plies, beads, tread, side wall and gum strips – their inspection Drum inspection for drumset, drum circumference Significance of parameters for tyre building. Size making on finished tyre and the relation to building specifications. Tyre building specifications sequence of building. Intermitant consolidation use of various cements and gum strips. Importance of the state of the Art Technology. Appraisal of Tyre building as most crucial operation correlation of some of the cured tyre& service returned tyres to the lack of building skill. Green tyre inspection procedures weight tolerance techno-commercial importance of green tyre weight. Green tyre storage considerations	7
5	GREEN TYRE PREPARATION & CURING : Internal and External painting – Awling – Bagging in case of Air bag cure Bag-o- matic and Air bag curing – mold lubrication- Bladder assembly bead curing rings – Dimension criticality Services to the Bag-o-matic presses Curing cycle – shaping – HPS, and hot water circulation. Dome steam cold water & vacuumcycles. Determination of optimum cure of tyres by thermos-couple built tyres. Economics of curing post cure inflation of Nylon tyres cured tyre inspection. Defects of tyres – Tyre classification for defects – causes and discussions - Examination of: (i) Returned tyres (ii) Tyres for retreading - Norm of tyre adjustments for fastwear, poor retreading Bead/casing failures. Hot and cold process retreading concept of total price/km run increasing competition and future trends in the industry and open house discussion.	7
Total		36
Course Outcomes:		
After completion of the course, students will be able to:		
1	To understand various components used and their function of tyres	
2	To design and suitable compounding formulation for various tyre components	
3	To know the building & curing of tyres	

Learning Resources:	
1	Tom French, Tyre technology, The University of Michigan, 1989.
2	Blow. C. M, Rubber Technology and Manufacture, Butterworth- Heinemann, London, 1982
3	Maurice Morton, “Rubber Technology”, Springer, 3rd edition, 1987.
4	Claude Hepburn, “Rubber Technology and Manufacture”, Third Edition, 2005
5	Kovac. F. J, “Tyre Technology”, Good Year Tire & Rubber Company, 1973.
6	Samuel K. Clark, Mechanics of pneumatic Tires, National Bureau of standards, Monograph, US Govt. printing office, 1971.



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Course Name:	Quantitative methods for decision making		
Course Code:	HM-HU604	Category:	HM
Semester:	6	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To familiarize the student with advanced quantitative approaches and mathematical optimization techniques used to address managerial and industrial problems.		
2	The student will gain insight into modeling and rational approaches to decision-making and their contribution to organizational effectiveness.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	An overview of quantitative methods and probability. Decision-making and quantitative techniques: Forecasting methods: qualitative method, Time series Analysis: simple and weighted moving average methods, exponential smoothing; regression analysis, forecasting errors, Decision analysis: Decision making under certainty, decision making under uncertainty, decision making under risk, decision trees, Job sequencing: Assumptions in sequencing problem, processing of n jobs through two machines	9
2	Linear programming problem: Formulation of linear programming problems, graphical method of solution, simplex method, artificial variable technique, duality in LPP, sensitivity analysis	8
3	Transportation Problem: Introduction, Formulation of Transportation Problem, the Initial Basic Feasible Solution, optimal solution. Assignment Problem: Introduction, Mathematical Formulation of the Problem, Hungarian Method	6
4	Project Management: Network construction, identification of critical path, various types of floats and their computations, Programme Evaluation and Review Technique (PERT) time calculations, crashing of network	5
5	Inventory models- Need and types of inventories, inventory associated costs, Economic order quantity, Classical EOQ inventory model with uniform demand rate and infinite replenishment. EOQ model with multiple price breaks, inventory classification systems	4
6	Queuing theory: Basis of Queuing theory, elements of queuing theory, Operating characteristics of a queuing system, Queue discipline, Service Mechanism, Classification of Queuing models, [M/M/1] Queue System	4
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Apply forecasting methods for predicting demands.
2	Make decisions under certainty, uncertainty, and conflicting situations.
3	Apply linear programming tools for optimal utilization of resources in various industries.
4	Solve transportation problems to minimize cost and understand the principles of assignment of jobs.
5	Understand the basic elements of a Queuing model and manage inventory with cost-effectiveness
6	Apply PERT/CPM for project scheduling and resource allocation in an optimal way

Learning Resources:	
1	Bedi K., Production and Operations Management, 3 rd Edition, Oxford University Press, 2013
2	Bernard W Taylor III, Introduction to Management Science, 11 th Edition, Pearson, 2013
3	Taha H.A., Operations Research, Pearson, 9th Edition, 2014.
4	Vohra N.D., Quantitative Techniques in Management, 4 th Edition, McGraw-Hill, 2010.
5	Heizer, J., Render, B. Munson C., Operations Management – Sustainability and Supply Chain Management, 12 th edition, Pearson, 2017
6	Sharma J.K., Operations Research –Theory and Application, 2 nd revised Edition, Macmillan Publishers, 2003.



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Course Name:	Automobile Engineering Lab II (Engine Testing and Pollution Measurement Lab)		
Course Code:	PC-AUE691	Category:	Professional Core courses
Semester:	Sixth	Credit:	1.5
L-T-P:	0-0-3	Pre-Requisites:	Engineering Thermodynamics.
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05

Course Objectives:

1	To introduce the learners with the need for automotive engine testing methods and their importance.
2	To understand the principles and performance characteristics of Engine.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Study the Valve Timing Diagram for Four Stroke Engine.	3
2	Study the Valve Timing Diagram for Two Stroke Engine.	3
3	Studying the components and working principle of an MPFI engine.	3
4	Performance test and energy balance on MPFI engine at different load conditions.	3
5	Performance test and energy balance on 2-Stroke Petrol engine at different load conditions.	3
6	Performance test and energy balance on 2-Stroke Diesel Engine at different load conditions.	3
7	Performance test and energy balance on 4-Stroke Petrol engine at different load conditions.	3
8	Performance test and energy balance on 4-Stroke Diesel Engine at different load conditions.	3
9	Morse test on petrol engine.	3
10	Determination of flash and fire point of fuels and lubricating oil.	3
11	Determination of calorific value of different types of fuel by Bomb calorimeter.	3
12	Measurement of pollutants emitted from the vehicle by gas analyzer/ Orsat apparatus/ smoke meter.	3
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	gain the experimental knowledge on the performance and testing of engines.
2	Test the fuel and oil properties.
3	Gain the knowledge on emission test of the IC engines.
4	Control the emission and use of different equipment to conduct performance test.

Learning Resources:

1	Ganesan V., Internal Combustion Engines, Tata McGraw Hill Co., Third Edition, 2007.
2	Babu, A.K., Automotive Engines, Khanna Publishing House, Delhi, 2018.
3	Obert E.F., Internal Combustion Engines and Air Pollution, Harper and Row Publication Inc. New York, 1973.



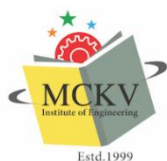
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Course Name:	Automobile Engineering Lab III (Automotive Design Lab)		
Course Code:	PC-AUE692	Category:	Professional Core courses
Semester:	Sixth	Credit:	1.5
L-T-P:	0-0-3	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05
Course Objectives:			
1	Acquire basic knowledge on automobile part designing using design software.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Sketcher: Introduction to CATIA/CREO, History, Basics, GUI, Use of mouse buttons, Sketcher, constraints, profile, setting workbench, Standard toolbar, how to open sketcher, sketch details and important toolbar for sketch, Profile toolbar.	3
2	Types of constraints, constraint application, constraint colour, Sketch constraint, view toolbar, Operation toolbar, Specification tree use, selecting toolbars, Sketch toolbar, Visualization toolbar. Toolbar setting, plane size setting, graphics properties toolbar.	3
3	Part Design: Introduction to Design tools like Extrude; Revolve; Shell; Pad etc needed to generate solid models using CATIA/CREO software.	3
4	Learning different tools of modeling software with exercise – Piston, Piston Pin,	6
5	Modelling of Connecting Rod.	6
6	Modelling of Bush big end, Bush small end,	3
7	Assembly Design: Assembly modeling of automotive mechanicals exercises – Piston- Connecting Rod – Crankshaft Assembly, Cam – Follower Assembly, Gear Assembly etc.	6
8	Spur Gear Design with formula	3
9	CNC Programming	3
Total		
Course Outcomes:		
After completion of the course, students will be able to:		
1	Construct parametric and feature model solid models	
2	Perform construction, analysis, and interrogation of Automobile parts models	
3	Build assembly models and fits	
4	Perform basic finite element analysis with CATIA/ CREO or similar 3D modeling software	

Learning Resources:	
1	Lab manual as per above experiments to be prepared by the institution



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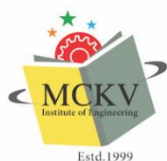
NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
 243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
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Course Name:	Automobile Engineering Lab IV (Vehicle Maintenance Lab)		
Course Code:	PC-AUE693	Category:	Professional Core Courses
Semester:	Sixth	Credit:	1.5
L-T-P:	0-0-3	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05
Course Objectives:			
1	Identify the problems with their causes and remedies by dismantling the given clutch, gear box differential assembly.		
2	Students are able to service the transaxle and rear axle assembly, brake system, identify the faults and service the same.		
3	Students are able to detect the fault in Anti-lock brake system.		
4	Students will be able to Service the steering gear boxes, and measure the turning circle radius and check wheel balancing and set wheel alignment parameters of a given vehicle.		
5	To align the head lamp of the given vehicle.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Inspection and servicing of different types of clutches.	3
2	Inspection and servicing of different types of gear boxes, transaxle and transfer case assembly.	3
3	Measurement of backlash and run out of differential unit and Servicing of different types of rear axle assembly.	3
4	Servicing of steering gear boxes and verification of Ackerman steering geometry.	3
5	Servicing and trouble-shooting of brake system, brake bleeding on a vehicle & study and fault detection of antilock braking system.	3
6	Head light beam alignment	3
7	Wheel balancing	3
8	Engine compression test	3
9	Testing of a nozzle	3
10	Tappet adjustment & valve timing diagram of four stroke engine	3
11	Inspection of tyre and tube	3
12	Wheel alignment.	3
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Service the types of clutches, gear boxes and calculation of torque carrying capacity and gear ratio
2	Service the transaxle and rear axle assembly, brake system, identify the faults and service the same
3	Service the steering gear boxes, and measure the turning circle radius and check wheel balancing and set wheel alignment parameters of a given vehicle.
4	Align the head lamp and wheel of the given vehicle
5	Know about the balancing the wheel, testing nozzle and compression testing

Learning Resources:	
1	Lab manual as per above experiments to be prepared by the institution



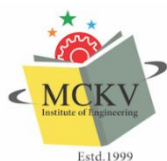
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Course Name:	Aptitude Skill Development-II		
Course Code:	MC671	Category:	Mandatory Course
Semester:	Six	Credit:	0
L-T-P:	2-0-0	Pre-Requisites:	Quantitative Ability, Logical and Verbal Reasoning
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To be prepared in the area of Quantitative Ability as well as Logical and Verbal Reasoning for Campus Placements and different Competitive Exams		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Verbal: Reading Comprehension, Para Jumbles, Email Writing, Resume Writing	3
2	Game based Cognitive Skills, Tournaments	3
3	Solve company-oriented campus placements aptitude papers covering Quantitative Ability, Logical Reasoning and Verbal Ability.	12
4	MCQ Based Strategies/Sort cuts and Mock test	6
Total		24
Course Outcomes:		
After completion of the course, students will be able to:		
1	Prepared for Campus Placements and different Competitive Exams	

Learning Resources:	
1	Arun Sharma,"Quantitative abilities", McGraw-Hill
2	R.S.Agrawal, "Quantitative Aptitude for Competitive Examinations", S. Chand
3	R.S.Agarwal,"A Modern Approach to Verbal & Non-Verbal Reasoning ",S.Chand



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Course Name:	Essence of Indian Knowledge Tradition		
Course Code:	MC673	Category:	Mandatory Course
Semester:	Sixth	Credit:	0
L-T-P:	2-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: NA	Continuous Assessment/Sessional: 100	Attendance: NA
Course Objectives:			
1	To impart basic principles of thought process, reasoning and inferencing. Sustainability is at the core of Indian Traditional knowledge Systems connecting society and nature		
2	To facilitate students with the concepts of Indian traditional knowledge and to make them understand the importance of the root of knowledge system		
3	To appreciate holistic life style of Yogic Science and wisdom capsules in Sanskrit Literature are also important in modern society with rapid technological advancements and social disruptions.		
4	The Course focuses on Introduction to Indian Knowledge System, Indian Perspective of modern scientific world-view and basic principles of Yoga and holistic health care system.		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction to traditional knowledge: Define traditional knowledge, nature and characteristics, scope and importance, kinds of traditional knowledge, the physical and social contexts in which traditional knowledge develop, the historical impact of social change on traditional knowledge systems. Indigenous Knowledge (IK), characteristics, traditional knowledge vis-à-vis indigenous knowledge, traditional knowledge Vs western knowledge traditional knowledge vis-à-vis formal knowledge	4
2	Basic structure of Indian Knowledge Systems (i) Veda (ii) Upa-Veda(iii) Vedanga (iv) Upanga; Modern Science and Indian Knowledge Systems Yoga and Holistic Health Care; Case Studies	4
3	Protection of traditional knowledge (TK): the need for protecting traditional knowledge Significance of TK Protection, value of TK in global economy, Role of Government to harness TK.	4
4	Legal frame work and TK: A: The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, Plant Varieties Protection and Farmer's Rights Act, 2001 (PPVFR Act); B: The Biological Diversity Act 2002 and Rules 2004, the protection of traditional knowledge bill, 2016. Geographical indicators act 2003.	4
5	Traditional knowledge and intellectual property: Systems of traditional knowledge protection, Legal concepts for the protection of traditional knowledge, Certain non IPR mechanisms of traditional knowledge protection, Patents and traditional knowledge.	4

6	Traditional knowledge in different sectors: Traditional knowledge and engineering, Traditional medicine system, TK and biotechnology, TK in agriculture, Importance of conservation and sustainable development of environment, Management of biodiversity, Food security of the country and protection of TK.	4
Total		24

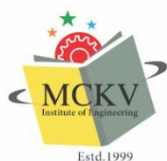
Course Outcomes:

After completion of the course, students will be able to:

1	Understand the concept of Traditional knowledge and its importance
2	Know the need and importance of protecting traditional knowledge.
3	Know the various enactments related to the protection of traditional knowledge.
4	Understand the concepts of Intellectual property to protect the traditional knowledge

Learning Resources:

1	A. Jha, Traditional Knowledge System in India, 2009.
2	B.K. Mohanta and V.K. Singh, Traditional Knowledge System and Technology in India, Pratibha Prakashan, 2012.
3	Shivaramakrishna V(Ed.) Cultural Heritage of India- Course Material, Bharatiya Vidya Bhavan, Mumbai
4	K. Kapoor and M. Danino, Knowledge Traditions and Practices of India, Central Board of Secondary Education, 2012.
5	Yoga Sutra of Patanjali, Ramakrishna Mission, Kolkata
6	E-Resources: http://nptel.ac.in/courses/121106003/



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Course Name:	Project-II (Summer Internship)		
Course Code:	PW-AUE671	Category:	Project/ Internships/ Sessional
Semester:	Sixth	Credit:	3
L-T-P:	0-0-0	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 100		
Course Objectives:			
1	To acquaint with the process of undertaking literature survey/industrial visit and identifying the problem		
2	To familiarize the process of solving the problem		
3	To acquaint with the process of applying basic engineering fundamental in the domain of practical application		

Course Contents:	
Description of Topic	Contact Hrs.
The project work could be done in the form of a summer project or internship in the industry or even a minor practical project in the college. Participation in any technical event/ competition to fabricate and demonstrate an innovative machine or product could be encouraged under this course.	90
Total	90
Course Outcomes:	
After completion of the course, students will be able to:	
1	understand the procedure to carry out practical projects related to any technical event/ competition to fabricate and demonstrate an innovative machine or product, etc.
2	Student is able to test the theoretical learning in practical situations by accomplishing the tasks assigned.